

Healthcare Trends, Innovation & Patient Care

Healthcare Data Analysis

Presented by Rishabh Arya



Introduction

This project focuses on analyzing healthcare data using MySQL. It explores patient demographics, medical conditions, hospital performance, admission trends, and billing patterns to generate meaningful insights and support better decision-making.



- Healthcare Data Analysis: Analyzing patient using MySQL.
- Identifying patient trends, medical conditions, and hospital performance.
- Analyzing admission patterns and billing trends to support better decision-making.

Overview



Patient Analysis

Analyze patients Demographics including age, gender, and blood group.



Hospital Analysis

Identify top-performing hospitals, admission trends, and medical conditions.



Billing Analysis

Examine billing amounts, medication records, and admission types.

The project uses SQL queries and data analysis techniques to identify key healthcare trends and generate meaningful insights.

Background



Past

Healthcare organizations have always generated large amounts of patient information, including medical records, admission details, treatments, and billing information. Managing these records efficiently is essential for understanding healthcare operations.



Present

Today, SQL and database technologies make it easier to organize, clean, and analyze healthcare data. This project uses MySQL to examine patient demographics, medical conditions, hospital performance, admission patterns, and billing records.



Future

The analysis aims to identify meaningful trends and patterns in healthcare data. These insights can help understand hospital performance, patient admission trends, and billing distribution, supporting better planning and decision-making.

Objective

01



Patient Demographics

To analyze patient information, including age, gender, and blood groups, and understand the distribution of patients across different categories.

02



Hospital Performance

To identify hospitals with the highest patient admissions and billing amounts and evaluate hospital performance.

03



Medical Condition Analysis

To examine common medical conditions, medication records, and test results to identify healthcare trends.

04



Billing & Admission Analysis

To analyze billing amounts, admission types, and patient stay durations to understand billing and admission patterns.

Key Insight

55500

Total Patients

Healthcare records analyzed.

141cr

Total Revenue

Total billing amount generated across all healthcare records.

16.77%

Arthritis

Percentage of records associated with the most frequently recorded condition.

Which month has the highest number of patient admissions, and how does admission volume change monthly

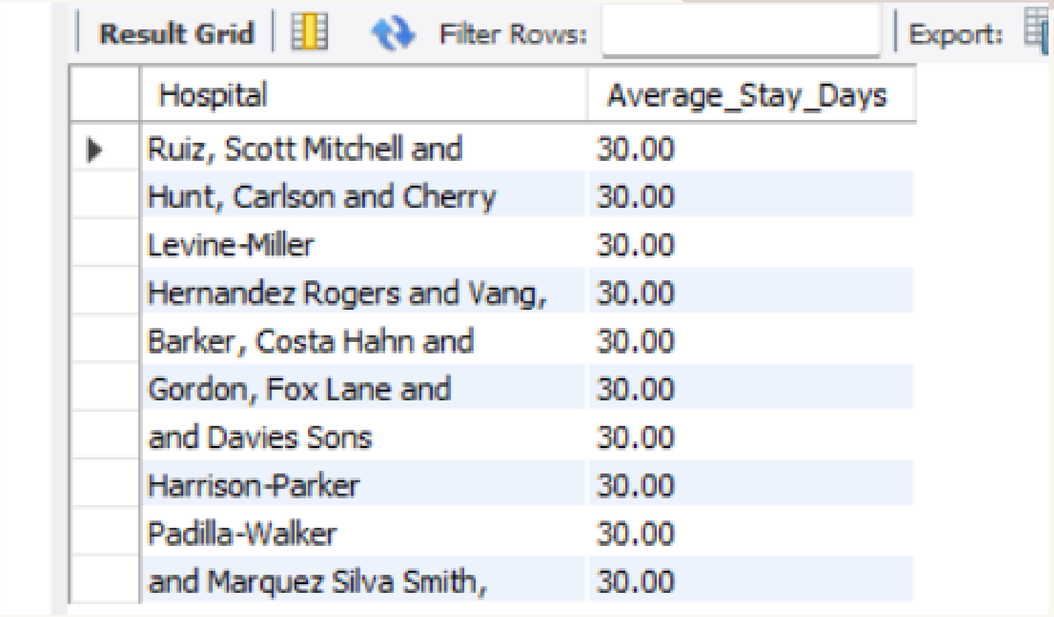


```
WITH Monthly_Admissions AS (
  SELECT
    YEAR(Date_of_Admission) AS Admission_Year,
    MONTH(Date_of_Admission) AS Admission_Month,
    COUNT(*) AS Total_Admissions
  FROM health
  GROUP BY
    YEAR(Date_of_Admission),
    MONTH(Date_of_Admission)),
Admission_Trend AS (
  SELECT *,
    LAG(Total_Admissions) OVER (
      ORDER BY Admission_Year, Admission_Month
    ) AS Previous_Month_Admissions
  FROM Monthly_Admissions)
SELECT *, Total_Admissions -
Previous_Month_Admissions
 AS Admission_Change
FROM Admission_Trend
ORDER BY Admission_Year, Admission_Month;
```

Which hospitals have the longest average patient stay

```
SELECT
   Hospital,
   ROUND(AVG(DATEDIFF(Discharge_Date,
Date_of_Admission)), 2)
      AS Average_Stay_Days
FROM health
WHERE Discharge_Date IS NOT NULL
  AND Date_of_Admission IS NOT NULL
  AND Discharge_Date >= Date_of_Admission
GROUP BY Hospital
ORDER BY Average_Stay_Days DESC
LIMIT 10;
```

Output



Result Grid | Filter Rows: | Export:

Hospital	Average_Stay_Days
Ruiz, Scott Mitchell and	30.00
Hunt, Carlson and Cherry	30.00
Levine-Miller	30.00
Hernandez Rogers and Vang,	30.00
Barker, Costa Hahn and	30.00
Gordon, Fox Lane and	30.00
and Davies Sons	30.00
Harrison-Parker	30.00
Padilla-Walker	30.00
and Marquez Silva Smith,	30.00

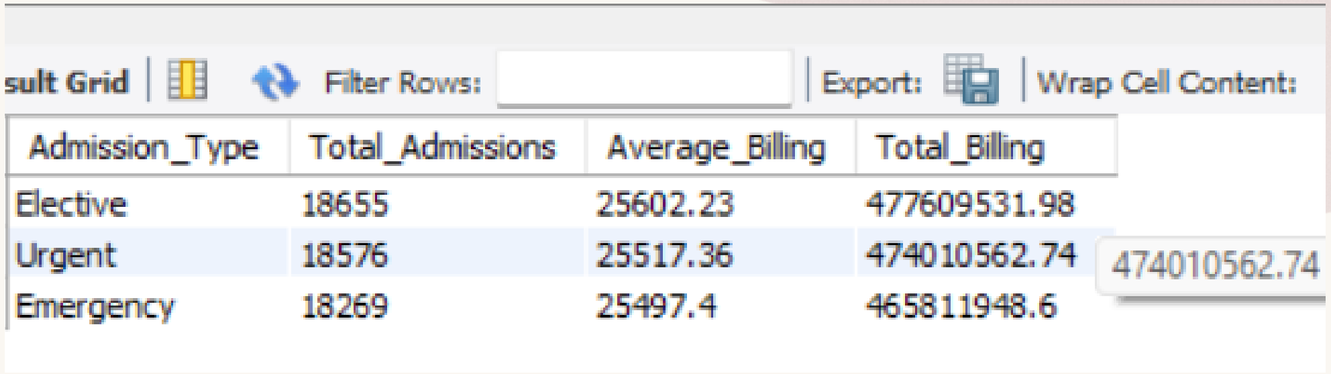
What are the top 3 most common medical conditions in each hospital

```
WITH Condition_Ranking AS (
  SELECT
    Hospital,
    Medical_Condition,
    COUNT(*) AS Total_Cases,
    RANK() OVER (
      PARTITION BY Hospital
      ORDER BY COUNT(*) DESC
    ) AS Condition_Rank
  FROM health
  GROUP BY Hospital, Medical_Condition
)
SELECT *
FROM Condition_Ranking
WHERE Condition_Rank <= 3
ORDER BY Hospital, Condition_Rank;
```

Hospital	Medical_Condition	Total_Cases	Condition_Rank
Abbott and Thompson, Sullivan	Hypertension	1	1
Abbott Inc	Arthritis	1	1
Abbott Ltd	Arthritis	1	1
Abbott Moore and Williams,	Obesity	1	1
Abbott-Castillo	Arthritis	1	1
Abbott-Coleman	Diabetes	1	1
Abbott-Ferrell	Asthma	1	1
Abbott-Hill	Obesity	1	1
Abbott-Jones	Hypertension	1	1
Abbott-Martin	Asthma	1	1
Abbott-Martin	Arthritis	1	1
Abbott-Rios	Diabetes	1	1
Abbott-Wilson	Obesity	1	1
Abbott, Peters and Hoffman	Diabetes	2	1
Abbott, Vazquez Bautista and	Obesity	1	1

Which admission type has the highest average billing amount?

```
SELECT
   Admission_Type,
   COUNT(*) AS Total_Admissions,
   ROUND(AVG(Billing_Amount), 2) AS
Average_Billing,
   ROUND(SUM(Billing_Amount), 2) AS
Total_Billing
FROM health
GROUP BY Admission_Type
ORDER BY Average_Billing DESC;
```

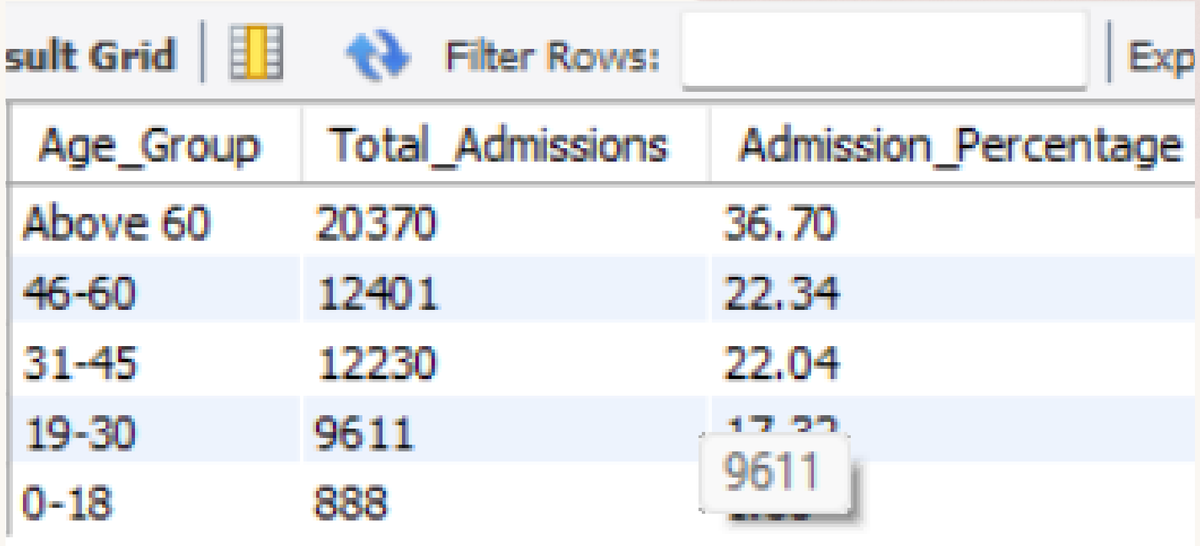


Admission_Type	Total_Admissions	Average_Billing	Total_Billing
Elective	18655	25602.23	477609531.98
Urgent	18576	25517.36	474010562.74
Emergency	18269	25497.4	465811948.6

Which age group has the highest number of hospital admissions?

```
SELECT
  CASE
    WHEN Age <= 18 THEN '0-18'
    WHEN Age <= 30 THEN '19-30'
    WHEN Age <= 45 THEN '31-45'
    WHEN Age <= 60 THEN '46-60'
    ELSE 'Above 60'
  END AS Age_Group,
COUNT(*) AS Total_Admissions,
ROUND(
    COUNT(*) * 100.0 / (SELECT COUNT(*) FROM
health),
    2
  ) AS Admission_Percentage

FROM health
GROUP BY Age_Group
ORDER BY Total_Admissions DESC;
```



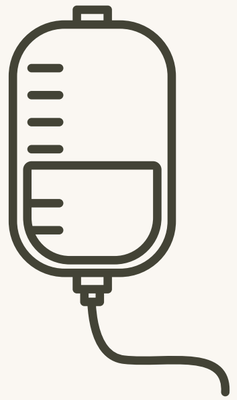
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Age_Group	Total_Admissions	Admission_Percentage
Above 60	20370	36.70
46-60	12401	22.34
31-45	12230	22.04
19-30	9611	[illegible]
0-18	888	[illegible]

9611

Future Outlook

This project provides a foundation for understanding healthcare data through SQL analysis. In the future, the analysis can be expanded by integrating Power BI dashboards for better visualization, automating data cleaning and reporting processes, and analyzing larger healthcare datasets. These improvements can help hospitals monitor patient trends, evaluate operational performance, and make more informed data-driven decisions.



Recommendations



01

Healthcare organizations should regularly clean and validate patient records to reduce duplicate entries, missing values, and incorrect information.

02

Hospitals should use Power BI dashboards to monitor patient admissions, billing patterns, medical conditions, and hospital performance.

03

Healthcare organizations should regularly analyze patient trends, admission patterns, and billing information to improve operational planning and resource allocation.

Conclusion

- Successfully analyzed 55,500 healthcare records using MySQL to identify meaningful patterns and trends.
- Explored patient demographics, medical conditions, hospital performance, admission trends, and billing patterns using SQL queries.
- Applied SQL concepts such as aggregate functions, CTEs, subqueries, and window functions to generate valuable healthcare insights.

Any question?



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Thank You

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